AMTI (NMTC) - 2017

BHASKARA CONTEST - JUNIOR LEVEL

PART - A

1. The sum of the values x, y that satisfy the equations $(x + y)2^{y-x} = 1$, $(x + y)^{x-y} = 2$ simultaneously is

(A) 2 (B) $\frac{3}{2}$ (C) $\frac{5}{2}$ (D) $\frac{7}{2}$

2. If $\left(2-\frac{a}{4}-\frac{4}{a}\right)\times\left\{(a-4)\sqrt[3]{(a-4)^{-3}}-\frac{(a^2-16)^{-1/2}(a-4)^{-1/2}}{(a+4)^{-3/2}}\right\}\times\left(\frac{a+4}{a-4}\right)=2016$ the value of a is

(A) $\frac{4}{1007}$ (B) $\frac{3}{2016}$ (C) $\frac{4}{2017}$ (D) none of these

3. ABCD is a square inscribed in a circle of radius 1 unit. The tangent to be circle at C meets AB produced at P. The length of PD is

(A) 2 (B) 3 (C) $\sqrt{13}$ (D) $\sqrt{10}$

4. Quadrilateral ABCD is inscribed in a circle with radius 1 unit. AC is the diameter of the circle and BD = AB. The diagonals cut at P. If PC = $\frac{2}{5}$ then the length of CD is equal to

(A) $\frac{2}{3}$ (B) $\frac{2}{7}$ (C) $\frac{1}{8}$ (D) $\frac{3}{4}$

5. The number of natural numbers n for which the expression $\frac{23n^2+18n+4}{n}$ is also a natural number is

(A) 3 (B) 2 (C) 1 (D) 0

6. The cost price of 16 oranges is equal to the selling price of 12 oranges. Then there is a

(A) 40% profit (B) 20% loss (C) $33\frac{1}{3}$% profit (D) $23\frac{1}{3}$% profit

7. The number of positive integer pairs (a, b) such that $ab - 24 = 2b$ is

(A) 6 (B) 7 (C) 8 (D) 9

8. $A = (2 + 1)(2^2 + 1)(2^4 + 1) ..... (2^{2016} + 1)$. The value of $(A + 1)^{1/2016}$ is

(A) 4 (B) 2016 (C) 2^{4032} (D)

9. The sum of two numbers a, b where a < b is 1215 and their H.C.F. is 81. The number of pairs of such pairs (a, b) is

(A) 1 (B) 2 (C) 3 (D) 4

10. The first Republic Day of India was celebrated on 26th January 1950. What was the day of the week on that date ?

(A) Tuesday (B) Wednesday (C) Thursday (D) Friday

11. The 12 numbers $a_1, a_2, ..., a_{12}$ are in arithmetical progression. The sum of all these numbers is 354. Let $P = a_2 + a_4 + ... + a_{12}$ and $Q = a_1 + a_3 + ... + a_{11}$. If the ratio P : Q is 32 : 37, the common difference of the progression is

(A) 2 (B) 3 (C) 4 (D) 5

12. A shopkeeper marks the prices of his goods at 20% higher than the original price. There is an increase in demand of the goods, and he further increases the price by 20%. The total profit % is

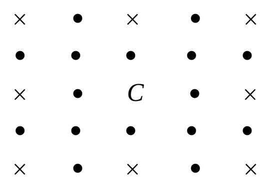
(A) 40 (B) 38 (C) 42 (D) 44

13. A circle passes through the vertices A and D and touches the side BC of a square ABCD. The side of the square is 2 cm. The radius of the circle (in cm) is

(A) $\frac{5}{4}$ (B) $\frac{4}{5}$ (C) 1 (D) $\frac{5}{2}$

14. There are four balls – one green, one red, one blue and one yellow and there are four boxes – one green, one red, one blue and one yellow. A child playing with the balls decides to put the balls in the boxes, one ball in each box. The number of ways in which the child can put the balls in the boxes such that no ball is in a box of its own color is
(A) 12 (B) 9 (C) 24 (D) 6

15. The 5 x 5 array of the dots represents trees in an orchard. If you were standing at the central spot marked C, you would not be able to see 8 of the 24 trees. (shown as X). If you were standing at the centre of a 9 x 9 array of trees, how many of the 80 trees would be hidden ?



(A) 40 (B) 32 (C) 36 (D) 44

PART - B

16. a and b are positive integers such that $a^2 + 2b = b^2 + 2a + 5$. The value of b is ____.

17. After full simplification, the value of the product

$$\left(\sqrt{2+\sqrt{3}}\right)\left(\sqrt{2+\sqrt{2+\sqrt{3}}}\right)\times\left(\sqrt{2+\sqrt{2+\sqrt{2+\sqrt{3}}}}\right)\left(\sqrt{2-\sqrt{2+\sqrt{2+\sqrt{3}}}}\right).$$

18. ABCD is a rectangle with AD = 1 and AB = 2. DFEB is also a rectangle. The area of DFEB is _______.

19. The two digit number whose units digit exceeds the tens digit by 2 and such that the product of the number and the sum of its digits is 144 is ______.

20. If $x = \frac{p}{q}$ where p, q are integers having no common divisors other than 1, satisfies

$$\sqrt{x+\sqrt{x}} - \sqrt{x-\sqrt{x}} = \frac{3}{2}\sqrt{\frac{x}{x+\sqrt{x}}}.$$

21. AE and BF are medians drawn to the legs of a right angled triangle ABC. The numerical value of $\frac{AE^2 + BF^2}{AB^2}$ is ______.

22. AB is a chord of a circle with center O. AB is produced to C such that BC = OA. CO is produced to E. The value of $\frac{\angle AOE}{\angle ACE}$ is _______.

23. The number of two digit numbers that are less than the sum of the squares of their digits by 11 and exceed twice the product of their digits by 5 is _______.

24. AB is a diameter of circle and CD is a parallel chord. P is any point in AB. The numerical value of $\frac{PC^2 + PD^2}{PA^2 + PB^2}$ is _______.

25. In the sequence 1, 2, 2, 4, 4, 4, 4, 8, 8, 8, 8, 8, 8, 8, 8, ... the 2016th term is 2^n. Then n = ______.

26. Each root of the equation $ax^2 + bx + c = 0$ is decreased by 1. The quadratic equation with these roots $x^2 + 4x + 1 = 0$. The numerical value of b + c is ______.

27. The number of integers n such that $\frac{n+2}{n^2+1} > \frac{1}{2}$ is _______.

28. P_1 and P_2 are two regular polygons. The number of sides of P_1 and P_2 respectively are in the ratio 3 : 2 and the respective interior angles are in the ratio 10 : 9. Then the sum of the number of sides of P_1 and P_2 is _______.

29. In triangle ABC, F and E are the mid points of AB and AC respectively. P is any point on the side BC. The ratio $\frac{Area\ of\ \Delta ABC}{Area\ of\ \Delta FPE}$ is ______.

30. x, y, z are distinct real numbers such that $x + \frac{1}{y} = y + \frac{1}{z} = z + \frac{1}{x}$. The value of $x^2y^2z^2$ is _____.

AMTI (NMTC) - 2017

GAUSS CONTEST - JUNIOR LEVEL

(Standard - IX & X)

Note :

1. Fill in the response sheet with your Name, Class and the institution through which you appear in the specified places.
2. Diagrams are only visual aids; they are NOT drawn to scale.
3. You are free to do rough work on separate sheets.
4. Duration of the test : 2 pm to 4 pm – 2 hours.

PART - A

Note :

- Only one of the choices A, B, C, D is correct for each question. Shade the alphabet of your choice in the response sheet. If you have any doubt in the method of answering, seek the guidance of the supervisor.
- For each correct response you get 1 mark. For each incorrect response you lose $\frac{1}{2}$ mark.

1. If m is a real number such that $m^2 + 1 = 3m$, the value of $\dfrac{2m^5 - 5m^4 - 2m^3 - 8m^2}{m^2 + 1}$ is

   (A) 1 (B) 2 (C) -1 (D) -2

2. Consider the equation $\dfrac{7x}{2} - a = \dfrac{5x}{3} + 9$. The least positive a for which the solution x to the equation is a positive integer is

   (A) 1 (B) 2 (C) 3 (D) 4

3. If x = 2017 and y = $\dfrac{1}{2017}$, the value of $\left\{\dfrac{x/y + 2}{x/y + 1} + \dfrac{x}{y}\right\} \div \left\{\dfrac{x}{y} + 2 - \dfrac{x/y}{x/y + 1}\right\}$ is

   (A) 2017 (B) 2017^2 (C) $\dfrac{1}{2017^2}$ (D) 1

4. The ratio of an interior angle of a regular pentagon to an exterior angle of a regular decagon is

   (A) 4 : 1 (B) 3 : 1 (C) 2 : 1 (D) 7 : 3

5. The smallest integer x which satisfies the inequality $\dfrac{x-5}{x^2 + 5x - 14} > 0$ is

   (A) -8 (B) -6 (C) 0 (D) 1

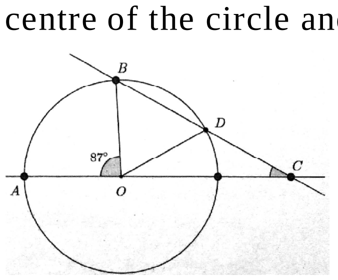
6. If x and y satisfy the equations $\sqrt{\dfrac{20y}{x}} = \sqrt{x+y} + \sqrt{x-y}, \sqrt{\dfrac{16x}{5y}} = \sqrt{x+y} - \sqrt{x-y}$ the value of $x^2 + y^2$ is

   (A) 2 (B) 16 (C) 25 (D) 41

7. 125% of a number x is y. What percentage of 8y is 5x ?

   (A) 30 % (B) 40 % (C) 50 % (D) 60 %

8. If the adjoining figure, O is the centre of the circle and OD = DC. If $\angle AOB = 87^0$, the measure of the angle $\angle OCD$ is

   (A) 27^0 (B) 28^0 (C) 29^0 (D) 19^0

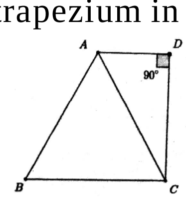
9. a, b, c, d, e are real numbers such that $\frac{a}{b}=\frac{2}{3}, \frac{b}{c}=\frac{1}{3}, \frac{c}{d}=\frac{1}{4}, e=\frac{ac}{b^2+c^2}$. The value of e is

(A) $\frac{1}{9}$ (B) $\frac{2}{9}$ (C) $\frac{1}{5}$ (D) $\frac{2}{5}$

10. The length and breadth of a rectangular field are integers. The area is numerically 9 more than the perimeter. The perimeter is

(A) 24 (B) 32 (C) 36 (D) 40

11. ABCD is a trapezium in which ABC is an equilateral triangle with area $9\sqrt{3}$ square units. If $\angle ADC = 90^0$, the area of the trapezium in square units is



(A) $12\sqrt{3}$ (B) $\frac{15\sqrt{3}}{2}$ (C) $\frac{27\sqrt{3}}{2}$ (D) $\frac{35\sqrt{3}}{2}$

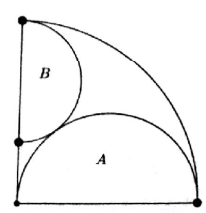
12. p is a prime number such that $p^2 - 8p - 65 > 0$. The smallest value of p is

(A) 7 (B) 11 (C) 13 (D) 17

13. The least positive integer n such that $2015^n + 2016^n + 2017^n$ is divisible by 10 is

(A) 1 (B) 3 (C) 4 (D) None of these

14. In a quadrant of a circle of diameter 4 units semicircles are drawn as shown. The radius of the smaller circle (B) is



(A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$

15. The product of two positive integers is twice their sum; the product is also equal to six times the difference between the two integers. The sum of these integers is

(A) 6 (B) 7 (C) 8 (D) 9

PART - B

Note :

- Write the correct answer in the space provided in the response sheet.
- For each correct response you get 1 mark. For each incorrect response you lose $\frac{1}{4}$ mark.

16. n is a natural number such that n minus 12 is the square of an integer and n plus 19 is the square of another integer. The value of n is _______.
17. The number of there digit numbers which have odd number of factors is _______.
18. The positive integers a, b, c are connected by the inequality $a^2 + b^2 + c^2 + 3 < ab + 3b + 2c$ then the value of $a + b + c$ is _______.
19. The sum of all roots of the equation $|3x - |1 - 2x|| = 2$ is _______.
20. PQR is a triangle with PQ = 15, QR = 25, RP = 30. A, B are points on PQ and PR respectively such that $\angle PBA = \angle PQR$. The perimeter of the triangle PAB is 28, then the length of AB is ________.
21. A hare sees a hound 100 m away from her and runs off in the opposite direction at a speed of 12 KM an hour. A minute later the hound perceives her and gives a chase at a speed of 16 KM an hour. The distance at which the hound catches the hare (in meters) is _____.
22. Two circles touch both the arms of an angle whose measure is 60^0. Both the circles also touch each other externally. The radius of the smaller circle is r. The radius of the bigger circle (in term of r) is _______.

23. a, b are distinct natural numbers such that $\frac{1}{a}+\frac{1}{b}=\frac{2}{5}$. If $\sqrt{a+b}=k\sqrt{2}$ the value of k is _______.

24. The side AB of an equilateral triangle ABC is produced to D such that BD = 2AC. The value of $\frac{CD^2}{AB^2}$ is _______.

25. D and E trisect the side BC of a triangle ABC. DF is drawn parallel to AB meeting AC at F. EG is drawn parallel to AC meeting AB at G. DF and EG cut at H. Then the numerical value of $\frac{Area(ABC)}{Area(DHE)+Area(AFHG)}$ is _______.

26. In an examination 70% of the candidates passed in English, 65 % passed in Mathematics, 27 % failed in both the subjects and 248 passed in both the subjects. The total number of candidates is ________.

27. In a potato race, a bucket is placed at the starting point, which is 7 m from the first potato. The other potatoes are placed 4 m a part in a straight line from the first one. There are n potatoes in the line. Each competitor starts from the bucket, picks up the nearest potato, runs back with it, drops in the bucket, runs back to pick up the next potato, runs to the bucket and drops it and this process continues till all the potatoes are picked up and dropped in the bucket. Each competitor ran a total of 150 m. The number of potatotes is ________.

28. A two digit number is obtained by either multiplying the sum of its digits by 8 and adding 1, or by multiplying the difference of its digits by 13 and adding 2. The number is _____.

29. The inradius of a right angled triangle whose legs have lengths 3 and 4 is _______.

30. a, b are positive reals such that $\frac{1}{a}+\frac{1}{b}=\frac{1}{a+b}$. If $\left(\frac{a}{b}\right)^3+\left(\frac{b}{a}\right)^3=2\sqrt{n}$, where n is a natural number, the value of n is ______.